

#This code will be used to demonstrate lemma 6.9. We will demonstrate that $\tilde{B}$ is positive along the γ curve.

#Construct the polynomial R from Theorem 6.1.3.

$$A := -(w + z)^3 \left((x^2 y^2 + z^4) w^4 - x^2 y^2 z^4 \right) :$$

$$B := x^2 y^5 w^2 z^2 (w + z)^2 (w - z) - x^2 y^2 (w + z)^3 (w^2 + z^2) w^2 z^2 (w - z) + w^4 y^3 z^4 (w + z)^3 + w^4 z^4 x^3 y^3 ;$$

$$R := B^2 \cdot ((w + z)^2 - y^2)^3 - A^2 \cdot (y^2 \cdot z \cdot w)^3$$

$$B := x^2 y^5 w^2 z^2 (w + z)^2 (w - z) - x^2 y^2 (w + z)^3 (w^2 + z^2) w^2 z^2 (w - z) + w^4 y^3 z^4 (w + z)^3 + w^4 z^4 x^3 y^3$$

$$R := (x^2 y^5 w^2 z^2 (w + z)^2 (w - z) - x^2 y^2 (w + z)^3 (w^2 + z^2) w^2 z^2 (w - z) + w^4 y^3 z^4 (w + z)^3 + w^4 z^4 x^3 y^3)^2 ((w + z)^2 - y^2)^3 - (w + z)^6 ((x^2 y^2 + z^4) w^4 - x^2 y^2 z^4)^2 y^6 z^3 w^3 \quad (1)$$

#Apply the rational parametrization constructed in this chapter 4..

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$y = a - \frac{1}{a} + b - \frac{1}{b},$$

$$x = a - \frac{1}{a} - b + \frac{1}{b} \Big\} :$$

simplify(numer(expand(subs(parametrization, R)))) ;

simplify(collect(numer(expand(subs(parametrization, B))), a));

$$\begin{aligned} & -(a + b)^{12} \left(-1024 b^{13} \left(b^2 - \frac{1}{2} \right) (b^2 + 1)^5 a^9 + 512 b^7 (b^{10} - 19 b^8 + 42 b^6 - 9 b^4 - 10 b^2 \right. \\ & \quad \left. + 1) (b^2 + 1)^4 a^{15} - 1536 b^9 \left(b^8 - b^6 + \frac{5}{3} b^4 - \frac{16}{3} b^2 + \frac{5}{3} \right) (b^2 + 1)^4 a^{13} \right. \\ & \quad \left. - 2560 b^{11} (b^2 + 1)^4 \left(b^6 - \frac{8}{5} b^4 + \frac{6}{5} b^2 - 1 \right) a^{11} + 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 \right. \\ & \quad \left. + 11 b^2 + 1) (b^2 + 1)^4 a^{17} + 5632 b^7 (b^2 + 1)^4 \left(b^6 - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11} \right) a^{21} \right. \\ & \quad \left. - 3584 b^7 (b^2 + 1)^4 \left(b^8 - \frac{52}{7} b^6 + \frac{95}{7} b^4 - \frac{43}{7} b^2 + \frac{3}{7} \right) a^{19} + b^{16} - 2560 (b^2 \right. \end{aligned}$$

$$\begin{aligned}
& -2) b^7 (b^2 + 1)^4 \left(b^2 - \frac{1}{5} \right) a^{23} + (-1016 b^{26} - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} \\
& - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} + 8 b^8 + 8 b^6) a^{10} + (-265 b^{24} \\
& - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} + 144 b^{12} - 72 b^{10} - 9 b^8) a^8 + (\\
& -512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} + 72 b^{12}) a^6 + (-250 b^{20} - 288 b^{18} \\
& + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} \\
& + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + (72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} \\
& + 72 b^{12} - 1024 b^{10}) a^{26} + (b^{32} + 8 b^{30} + 20 b^{28} - 120 b^{26} - 3426 b^{24} - 7288 b^{22} \\
& - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} - 157780 b^{12} - 32376 b^{10} + 1950 b^8 \\
& + 1672 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + a^{32} b^{16} - 1024 b^{15} (b^2 + 1)^4 a^7 + 2 b^4 (b^{24} + 36 b^{22} \\
& - 2636 b^{20} - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^8 \\
& - 1372 b^6 + 180 b^4 + 36 b^2 + 1) a^{12} - 4 b^2 (b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} \\
& + 34120 b^{18} + 65514 b^{16} + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^8 + 378 b^6 + 37 b^4 \\
& + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} + 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} \\
& + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} + 36360 b^{10} + 5211 b^8 - 2118 b^6 - 539 b^4 + 76 b^2 \\
& + 1) a^{18} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} \\
& - 84792 b^{10} - 12876 b^8 - 348 b^6 + 3252 b^4 + 420 b^2 - 127) a^{20} + 8 \left(b^{20} + b^{18} - 27 b^{16} \right. \\
& \left. - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^8 + 2369 b^6 - 699 b^4 + b^2 + 1 \right) b^6 a^{22} \\
& - 9 b^8 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^8 - 328 b^6 - 528 b^4 + \frac{4168}{9} b^2 + 1 \right) a^{24} \Big) \\
& (b^2 + 1)^3 (a^2 + 1)^3 (ab + 1)^6 (ab - 1)^6 \\
& 2 (a + b)^3 (b^2 + 1)^2 (a^2 + 1)^2 \left((b^5 - 2 b^3) a^{10} + (-2 b^7 + 6 b^5 - 3 b^3) a^8 + 5 (b^2 + 1)^3 \left(b^2 \right. \right. \textbf{(2)}
\end{aligned}$$

$$\begin{aligned} & -\frac{1}{5} \Big) a^7 + (b^9 - 6b^7 + 7b^5) a^6 + (-b^{10} + 6b^6 + 8b^4 + 3b^2) a^5 + (2b^9 - 5b^7 \\ & + b^3) a^4 + 2b^4 (b^2 + 1)^3 a^3 + (b^9 - 2b^5) a^2 + b^7 \Big) (ab + 1)^3 (ab - 1)^3 \end{aligned}$$

#Define r1

$$\begin{aligned} r1 := & \left(b^{16} - 4b^2 (b^{28} + 12b^{26} - 155b^{24} + 1018b^{22} + 4315b^{20} + 34120b^{18} + 65514b^{16} \right. \\ & + 57358b^{14} + 41514b^{12} + 19656b^{10} + 4187b^8 + 378b^6 + 37b^4 + 12b^2 + 1) a^{14} + 2b^4 (b^{24} \\ & + 36b^{22} - 2636b^{20} - 6620b^{18} - 27212b^{16} - 64184b^{14} - 54798b^{12} - 25400b^{10} - 9036b^8 \\ & - 1372b^6 + 180b^4 + 36b^2 + 1) a^{12} - 1024b^{15} (b^2 + 1)^4 a^7 - 4b^2 (b^{28} + 12b^{26} + 37b^{24} \\ & - 518b^{22} - 2085b^{20} + 17992b^{18} + 38058b^{16} + 78414b^{14} + 92394b^{12} + 36360b^{10} + 5211b^8 \\ & - 2118b^6 - 539b^4 + 76b^2 + 1) a^{18} + 2b^4 (b^{24} + 36b^{22} + 180b^{20} - 2396b^{18} - 25164b^{16} \\ & - 42552b^{14} - 90894b^{12} - 84792b^{10} - 12876b^8 - 348b^6 + 3252b^4 + 420b^2 - 127) a^{20} \\ & + 8b^6 \left(b^{20} + b^{18} - 27b^{16} - 159b^{14} + \frac{2163}{2}b^{12} - 1827b^{10} - \frac{1037}{2}b^8 + 2369b^6 - 699b^4 \right. \\ & + b^2 + 1 \Big) a^{22} - 9b^8 \left(b^{16} + 8b^{14} - 16b^{12} - 72b^{10} + \frac{3698}{9}b^8 - 328b^6 - 528b^4 + \frac{4168}{9}b^2 \right. \\ & + 1 \Big) a^{24} + (-4b^{18} + 8b^{16} - 4b^{14}) a^{30} + (6b^{20} - 32b^{18} + 44b^{16} - 32b^{14} + 6b^{12}) a^{28} \\ & + (72b^{20} - 76b^{18} + 8b^{16} + 692b^{14} + 72b^{12} - 1024b^{10}) a^{26} + (b^{32} + 8b^{30} + 20b^{28} \\ & - 120b^{26} - 3426b^{24} - 7288b^{22} - 125524b^{20} - 222104b^{18} - 279672b^{16} - 317080b^{14} \\ & - 157780b^{12} - 32376b^{10} + 1950b^8 + 1672b^6 + 20b^4 + 8b^2 + 1) a^{16} + (-1016b^{26} \\ & - 8440b^{24} - 21976b^{22} - 25848b^{20} - 37428b^{18} - 31000b^{16} - 8756b^{14} + 264b^{12} - 216b^{10} \\ & + 8b^8 + 8b^6) a^{10} + (-265b^{24} - 1864b^{22} + 2192b^{20} + 4232b^{18} + 142b^{16} - 1912b^{14} \\ & + 144b^{12} - 72b^{10} - 9b^8) a^8 + (-512b^{22} - 696b^{20} + 692b^{18} + 1288b^{16} - 76b^{14} \\ & + 72b^{12}) a^6 + (-250b^{20} - 288b^{18} + 44b^{16} - 32b^{14} + 6b^{12}) a^4 + (-4b^{18} + 8b^{16} \\ & - 4b^{14}) a^2 + a^{32} b^{16} - 2560 \left(b^2 - \frac{1}{5} \right) b^7 (b^2 + 1)^4 (b^2 - 2) a^{23} - 1536 \left(b^8 - b^6 + \frac{5}{3}b^4 \right. \\ & \left. - \frac{16}{3}b^2 + \frac{5}{3} \right) b^9 (b^2 + 1)^4 a^{13} - 2560 \left(b^6 - \frac{8}{5}b^4 + \frac{6}{5}b^2 - 1 \right) b^{11} (b^2 + 1)^4 a^{11} \end{aligned}$$

$$\begin{aligned}
& - 1024 b^{13} (b^2 + 1)^5 \left(b^2 - \frac{1}{2} \right) a^9 + 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 + 11 b^2 \\
& + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 19 b^8 + 42 b^6 - 9 b^4 - 10 b^2 + 1) (b^2 + 1)^4 a^{15} \\
& + 5632 b^7 (b^2 + 1)^4 \left(b^6 - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11} \right) a^{21} - 3584 \left(b^8 - \frac{52}{7} b^6 + \frac{95}{7} b^4 \right. \\
& \left. - \frac{43}{7} b^2 + \frac{3}{7} \right) b^7 (b^2 + 1)^4 a^{19} \Big):
\end{aligned}$$

#Define $\tilde{B}$

$$\begin{aligned}
\tilde{B} := & \left((b^5 - 2 b^3) a^{10} + (-2 b^7 + 6 b^5 - 3 b^3) a^8 + 5 (b^2 + 1)^3 \left(b^2 - \frac{1}{5} \right) a^7 + (b^9 - 6 b^7 \right. \\
& + 7 b^5) a^6 + (-b^{10} + 6 b^6 + 8 b^4 + 3 b^2) a^5 + (2 b^9 - 5 b^7 + b^3) a^4 + 2 b^4 (b^2 + 1)^3 a^3 \\
& \left. + (b^9 - 2 b^5) a^2 + b^7 \right):
\end{aligned}$$

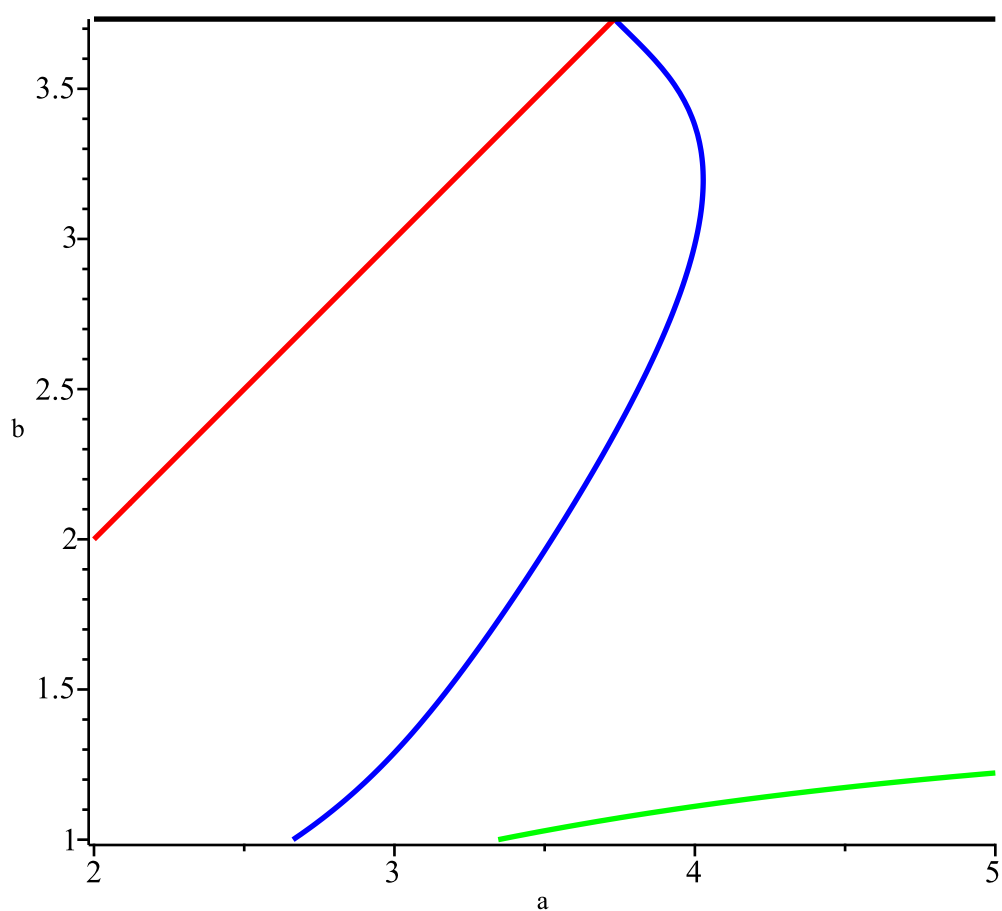
#Plot the curve $r_1=0$ and $\tilde{B}$ in region A1.

$r := a - b :$
 $rr := b - 2 - \text{sqrt}(3) :$

with(plots) :

```
implicitplot([r1=0,  $\tilde{B}=0$ , r=0, rr=0], a=2 .. 5, b=1 .. (2 + sqrt(3)),  
  scaling=constrained,  
  gridrefine=2,  
  thickness=2,  
  color=[blue, green, red, black],  
  labels=["a", "b"],  
  title="Graphic for  $r_1=0$  and  $\tilde{B}=0$  in A1");
```

Graphic for $r_1=0$ and $\tilde{B}=0$ in A1



#Calculate the resultant of r1 with respect to $\tilde{B}$ and respect to a.

$R1 := resultant(r1, \tilde{B}, a) :$

#Compute the factors of the resultant R1 and count how many there are

$bfactor := [op(factor(R1))]:$

$nops(bfactor);$

4

(3)

$bfactor[1]; bfactor[2]; bfactor[4]$

279841

b^{160}

$(b^2 + 1)^{128}$

(4)

$bfactor[3]$

$(374017 b^{48} - 1082256 b^{46} - 2821284 b^{44} + 38273024 b^{42} - 137706774 b^{40} - 25011408 b^{38}$
 $+ 2222925500 b^{36} - 11137697376 b^{34} + 34666181511 b^{32} - 79729549344 b^{30}$
 $+ 140202896376 b^{28} - 190367984256 b^{26} + 204746625068 b^{24} - 177535429920 b^{22}$
 $+ 126486144024 b^{20} - 74566977600 b^{18} + 36572797215 b^{16} - 14813913168 b^{14}$
 $+ 4932365356 b^{12} - 1327146624 b^{10} + 292102794 b^8 - 53430800 b^6 + 8765868 b^4$
 $- 983136 b^2 + 45241)^2$

(5)

$sturm(sturmseq(bfactor[3], b), b, 1, (2 + \sqrt{3})); \#number of roots in [1, (2 + \sqrt{3}))$

0

(6)

#Check the sign of $\tilde{B}$ at the point $(\gamma(1), 1)$

#Locate a range for $\gamma(1)$

sturm(sturmseq(eval(r_1 , $b = 1$), a), $a, 2, 3$); #number of roots in $[2, 3)$

1

(7)

#Verify that in this interval $\tilde{B}(a, 1)$ has a constant sign.

sturm(sturmseq(eval($\tilde{B}$, $b = 1$), a), $a, 2, 3$); #number of roots in $[2, 3)$

0

(8)

#Calculate the sign of $\tilde{B}(\gamma(1), 1)$ using the sign of $\tilde{B}(2, 1)$.

eval(eval($\tilde{B}$, $b = 1$), $a = 2$)

4061

(9)